

Yatri Shield: Parallel Testing Guide (Mobile & Web)

This guide is designed for a multi-person QA team to test the Yatri Shield ecosystem concurrently and verify end-to-end real-time synchronization.

- **Tester A** will operate the **Tourist Mobile App** (Physical Android device or Emulator).
- **Tester B** will operate the **Police/Responder Web Dashboard** (PC Browser).

Workflow 1: End-to-End Emergency SOS & Dispatch

Scenario: A tourist triggers a legitimate SOS. The responder acknowledges, dispatches a unit, and resolves the case in person using OTP verification.

Step	Tester A (Tourist Mobile App)	Tester B (Responder Web Dashboard)	Expected Outcome
1	Open the app, tap the SOS EMERGENCY button, and hold for 3 seconds.	Log in to /login/responder . Keep the Dashboard open.	Tester A: Siren plays, countdown finishes, screen turns red (Active SOS).
2	Observe the active tracking screen.	Monitor the Live Incidents Map & Table .	Tester B: Dashboard instantly flashes a new incident row with live GPS coordinates, battery, and altitude.
3	Wait for responder acknowledgment.	Click "Acknowledge & Dispatch" on the new incident row.	Tester B: Status changes to INVESTIGATING . Tester A: Receives push/socket update: <i>"Rescue is en route."</i>
4	Wait for the rescue officer to arrive (Simulated).	Click "Verify & Resolve" on the incident.	Tester B: System prompts for a 4-digit OTP to prevent false closures.
5	Read the 4-digit Rescue OTP displayed on the SOS screen to Tester B.	Enter the OTP provided by Tester A and click Confirm .	Tester B: Incident is marked RESOLVED . Tester A: SOS tracking terminates, screen shows "Incident Resolved," and redirects to Home.

Workflow 2: Safe PIN Cancellation (False Alarm)

Scenario: A tourist accidentally triggers the SOS and cancels it before police dispatch.

Step	Tester A (Tourist Mobile App)	Tester B (Responder Web Dashboard)	Expected Outcome
1	Trigger SOS EMERGENCY and let it activate.	Monitor Live Incidents Table .	Tester B: New SOS appears on the dashboard.
2	Immediately tap "Cancel SOS with Safe PIN" .	Watch the corresponding incident row.	Tester A: PIN prompt appears.
3	Enter the 4-digit PIN and confirm.	Keep watching the incident row.	Tester A: Tracking stops, app shows "SOS Cancelled", redirects to /home . Tester B: Incident row automatically updates to CANCELLED_BY_USER and is cleared from active emergencies.

 Workflow 3: Geofence Danger Zones & Offline Map Alerts

Scenario: Police authorities designate a high-risk area. The tourist's app dynamically updates and warns them.

Step	Tester A (Tourist Mobile App)	Tester B (Responder Web Dashboard)	Expected Outcome
1	Keep the app open on the Home or Trips map screen.	Navigate to Zones/Geofences (/responder/zones). Click Create Zone .	Tester B: Zone editor map opens.
2	Wait for synchronization.	Draw a polygon around Tester A's current GPS location .	Tester B: Polygon is defined.
3	Monitor the screen.	Add a Critical severity crime record to force the Area Safety Score below 50. Click Save & Broadcast .	Tester B: New zone is broadcasted via WebSockets/API.
4	Observe the map and notifications.	Verify the zone appears in the active list.	Tester A: App fetches the new zone in the background. The map instantly highlights the area in Crimson Red , and a High-Risk Warning banner appears on the Home screen.

 Workflow 4: Offline Edge AI Guidance (Zero Internet)

Scenario: A tourist loses cellular service in a remote area and needs urgent survival instructions.

Step	Tester A (Tourist Mobile App)	Expected Outcome
1	Turn on Airplane Mode (disable Wi-Fi & Cellular Data).	App status indicator changes to OFFLINE .
2	Navigate to the AI Guidance tab (/emergency-ai).	Chat interface loads from local storage.
3	Tap the quick prompt " Wild animals nearby " or type a query.	The offline Edge AI engine processes the request instantly (Zero Latency).
4	Read/Listen to the response.	The AI provides step-by-step triage. Text-to-Speech (TTS) automatically reads the instructions out loud.

 Workflow 5: Trip Creation & Automated Escalation

Scenario: A tourist begins a monitored high-altitude trek but misses their safety check-in.

Step	Tester A (Tourist Mobile App)	Tester B (Responder Web Dashboard)	Expected Outcome
1	Go to Trips . Click Start New Trip . Select "High-Altitude Trek" (Tier 2).	Go to the Active Trips / Tourists dashboard.	Tester A: Persistent GPS tracking begins. Check-in timer starts. Tester B: New active trip appears on the monitor.
2	Force a timeout (Simulate by jumping device time forward by 4 hours).	Monitor for automated escalation alerts.	Tester A: Phone vibrates with "Check-in Overdue" warning. Tester B: Trip is flagged OVERDUE in red, triggering SDRF alerts.
3	Open the app and tap " I'm OK (Check-in) ".	Watch the trip status.	Tester A: Timer resets. Tester B: Status reverts to ACTIVE (Green).

 Workflow 6: Tourist KYC & Identity Verification

Scenario: Ensuring tourist identities are cryptographically bound to emergency dispatches.

Step	Tester A (Tourist Mobile App)	Tester B (Responder Web Dashboard)	Expected Outcome
1	Complete initial onboarding. Enter Passport/Aadhaar details in KYC Screen .	(Not involved)	Tester A: App generates an offline-verifiable Decentralized ID (DID) and animated QR Code.

Step	Tester A (Tourist Mobile App)	Tester B (Responder Web Dashboard)	Expected Outcome
2	Trigger an SOS.	View the Incident Details modal.	Tester B: The dashboard securely decrypts and displays the verified KYC identity (Name, Age, Blood Type) attached to the SOS signal.

Generated for the Yatri Shield Quality Assurance Team.